Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*
Code: 100301 *Subject: Biology for Engineers* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Where does the light reaction takes place?
(i) Grana. (ii) Stroma
(iii) Cytoplasm. (iv). Endoplasmic reticulum

(b) COVID-19 is a
(i) Bacteria (ii) Protozoa (iii) Virus (iv) Fungus

(c) In the exothermic reaction, the enthalpy of reaction is always
(i) zero. (ii) positive. (iii) negative. (iv) none of these

(d) Photosynthesis is a
(i) exothermic reaction (ii) endothermic reaction
(iii) light reaction (iv) dark reaction

(e) Genetic engineering is related to
(i) plasmid (ii) mitochondria
(iii) ribosome (iv) golgi complex

(f) The simplest amino acid is
(i) glycine. (ii) alanine
(iii) asparagine. (iv) tyrosine

(g) Which of the following characteristics of pea plants was not used by Mendel in his experiments
(i) seed colour. (ii) seed shape
(iii) pod length. (iv) flower position

(h) The green colour of plant is due to _____
(i) Heparin (ii) Haeme
(iii) Chlorophyll. (iv) Hepatin

(i) Which of the following purine bases is present in RNA
(i) uracil (ii) thyamine (iii) cytosine (iv) guanine

(j) Chlorophyll is soluble in water
(i) True (ii) False

Q.2 (a) What is the need of study of Biology for an engineer? **[7]**
(b) Compare the working fundamentals of human eye and camera. **[7]**

Q.3 Write the answer in very brief: **[3½ x 4=14]**
(a) Write the difference between unicellular and multicellular organisms.
(b) Differentiate between plant cell and animal cell.
(c) List the laws of inheritance.
(d) Define molecular taxonomy of living organism.

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Q.4	(a)	Describe the Mendel's laws of segregation with suitable example.	[7]
	(b)	Discuss about law of dominance and law of independent assortment.	[7]
Q.5	(a)	What do you understand by Growth Kinetics? Explain its different phases.	[7]
	(b)	Define amino acids. Write briefly about different types of amino acids.	[7]
Q.6	(a)	Explain of enzyme action, with examples.	[7]
	(b)	Explain monitoring of enzyme catalysed reactions.	[7]
Q.7	(a)	What is the difference between RNA and DNA.	[7]
	(b)	How is genetic information stored and transferred from generation to generation.	[7]

Q.8 Write notes on any two of the following : [7x2=14]

(a) Genetic disorder disease in human being

(b) Role of glycolysis in metabolism Synthesis of glucose by photosynthesis in plant

Q.9	(a)	Describe fundamentals of thermodynamics with emphasis of free energy concept.	[7]
	(b)	Differentiate between endothermic reaction and exothermic reaction	[7]

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